18 Sept Wednesday Lab Assignment

1. Write a program in C to calculate the power of any number using recursion.
   example :
   Input the base value : 2
   Input the value of power : 6
   Expected Output :
   The value of 2 to the power of 6 is : 64

2. Write a recursive function in C programming to calculate sum of digits of a number
   Example
   Input
   Input number: 1234
   Output
   Sum of digits: 10

3. Write a C program to find GCD (HCF) of two numbers using recursion.
   Example
   Input
   Input first number: 10
   Input second number: 15
   Output
   HCF of 10 and 15 = 5

4. Write a program in C to find the sum of all elements of the array.
   Test Data :
   Input the number of elements to be stored in the array :3
   Input 3 elements in the array :
   element - 0 : 2
   element - 1 : 5
   element - 2 : 8
   Expected Output :
   Sum of all elements stored in the array is : 15

5. Write a program in C to count the total number of duplicate elements in an array.
   Test Data :
   Input the number of elements to be stored in the array :3
   Input 3 elements in the array :
   element - 0 : 5
   element - 1 : 1
   element - 2 : 1
   Expected Output :
   Total number of duplicate elements found in the array is : 1